NCERT Summary: Topic 16: Drainage Systems of India — Raw Notes

Source Integration

- NCERT Class 11 — India: Physical Environment, Chapter 3: Drainage System

Core Factual & Conceptual Architecture

1. Drainage Concepts & Patterns

- Definitions: Drainage refers to water flow through defined channels, forming a drainage system. Catchment areas collect water, drainage basins cover entire river networks, and watersheds separate adjacent basins.
- Drainage Patterns: Dendritic (tree-like, Northern Plain), radial (outward from a hill like Amarkantak), trellis (parallel primary with right-angle tributaries), and centripetal (converging into a depression).
- Drainage Classification: Divided by orientation into Bay of Bengal (~77 percent share) and Arabian Sea (~23 percent share); by watershed size into major (14 basins), medium (44 basins), and minor basins; and by origin into Himalayan and Peninsular drainage.

2. Himalayan Drainage Evolution & Systems

- Evolution: Evolved from the ancient Mio-Pleistocene Indo-Brahma (Shiwalik) river, later dismembered by tectonic upheavals and the formation of the Malda gap.
- Indus System: Originates near Bokhar Chu in Tibet (Singi Khamban), flowing northwest through Ladakh before entering Pakistan. Major tributaries include the Zaskar, Shyok, and the Panjnad (Jhelum, Chenab, Ravi, Beas, Satluj).
- Ganga System: Rises at Gangotri as Bhagirathi, joining Alaknanda at Devprayag. Features major Himalayan tributaries (Ramganga, Gomati, Ghaghara, Gandak, Kosi, Mahananda) and peninsular tributaries (Yamuna, Son).
- Brahmaputra System: Originates near Chemayungdung glacier in Tibet as Tsangpo, entering Arunachal as Dihang before flowing through Assam as the Brahmaputra and entering Bangladesh as Jamuna.

3. Peninsular Drainage System

- Characteristics: Older than Himalayan systems with graded shallow valleys, fixed courses, absence of meanders, and non-perennial flow.
- Evolution: Shaped by early Tertiary subsidence of the western flank, northern trough faulting creating Narmada and Tapi rift valleys, and slight northwest-southeast block tilting.
- Major Rivers: Mahanadi, Godavari (Dakshin Ganga, largest peninsular river), Krishna, Kaveri (fed by both monsoons), Narmada, and Tapi.

4. Water Resource Management & Challenges

- Distribution & Usability: Uneven temporal and spatial water distribution leads to simultaneous floods and droughts. Inter-basin water transfer is proposed as a partial remedy.
- Pollution & Cleanup: River water faces severe pollution from urban sewage, industrial effluents, and religious immersions. Major mitigation initiatives include Namami Gange and Ganga Action Plan.

✍ Notes & Updates